

"Hi Kids, Let's Talk About How Snakes Hunt": Understanding the Process of Children's Instructional Video Creation through a Workshop Study

Zhenyao Cai
School of Education
University of California Irvine
Irvine, California, USA
zhenyaoc@uci.edu

Ariel Han
University of Southern California
Los Angeles, California, USA
arielhan@usc.edu

Shiyao Wei
Florida State University
Tallahassee, Florida, USA
sw22b@fsu.edu

Kylie A Peppler
School of Education / Creativity Labs
University of California, Irvine
Irvine, California, USA
kpeppler@gmail.com

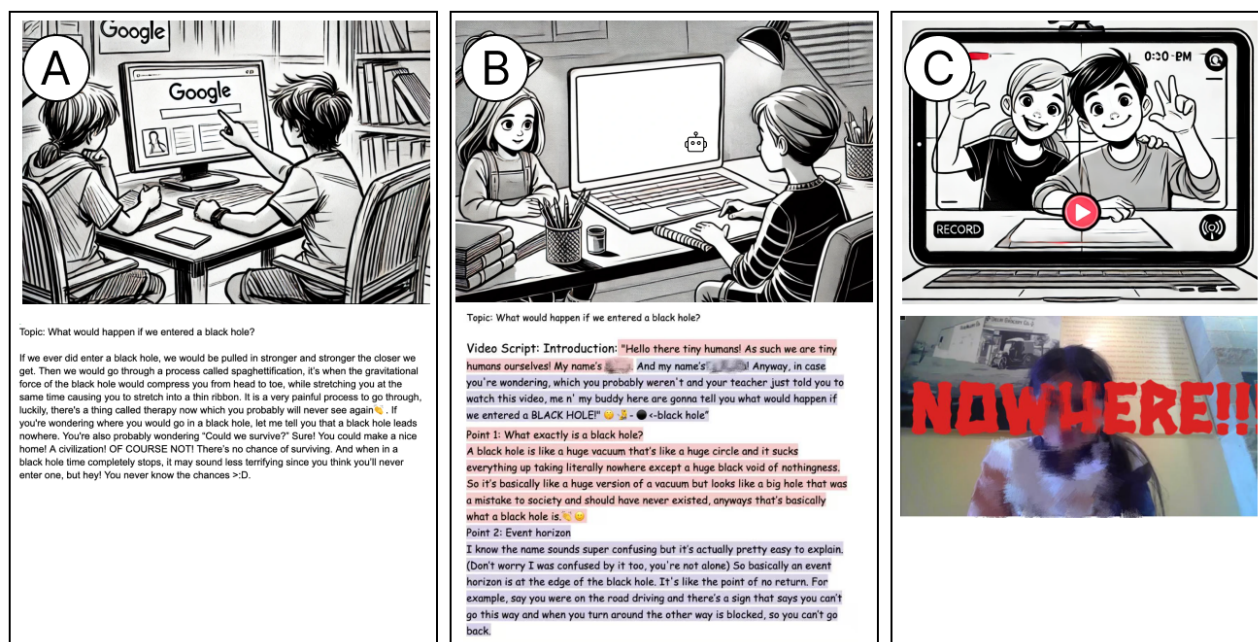


Figure 1: (A) Children use search engines to find relevant information about the topic. (B) Children organize information and write scripts with AI assistance. (C) Children record and edit videos.

Abstract

As children increasingly transition from media consumers to "prosumers" on user-generated content platforms, developing their



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media and content creation literacies becomes critical. Yet little research examines how children engage with video creation or how emerging technologies, such as generative AI, can support this creative process. This study explores children's engagement and challenges during a two-week, project-based learning workshop where they learned to create educational videos teaching school-aligned science topics. Our exploratory findings suggest that educational video creation activity, when designed properly, can serve as an intervention for the dual learning of science literacy

and media literacy. In the workshop, children showed different engagement profiles: some were self-directed and motivated by both science learning and video creation, some focused primarily on learning the science content, some treated the activity as a classroom assignment, and others disengaged from the process. Building on insights from our findings, We also extend Shneiderman's Collect-Relate-Create-Donate (CRCD) model to better support children's creativity. We propose design directions including conversational, filtered, and multimodal information gathering for the Collect phase; role-based peer and family interaction for the Relate phase; low-floor, high-ceiling creation tools for the Create phase; and positioning children as meaningful sharers of content while supporting education around content sharing, privacy, and audience awareness for the Donate phase.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; *Empirical studies in collaborative and social computing*; *Collaborative interaction*.

Keywords

Media Literacy, Content Creation, Creativity Support Tools, Child-AI Co-Creation

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1 Introduction

Media literacy is one of the essential 21st-century skills children need to develop, according to the OECD educational framework [14, 15]. With increasing access to the Internet and social media, teaching children to interact with the media critically has become more important. Media literacy is not only about teaching how to consume media, but also involves teaching how to create content more responsibly [30]. Although most research focuses on improving children's skills in critical media consumption, the creation and sharing of media content by children, or "*prosuming*," has been less explored.

Among the works of research on children's content creation, most focused on children creating media for self-expression, play, or curation of life [13, 56, 84], and often examined video types such as pretend play and acting [49] or personal vlogs about their lives [55, 84]. An intriguing yet underexplored type of content that children create and share is educational videos, where they teach concepts, processes, or ideas to a broader audience. A recent study by Cai and Wei [13] highlights that children can effectively deliver meaningful content through educational videos, such as teaching a science experiment or explaining a concept. However, despite these insights, there is limited research on the process of children creating videos and how creativity support tools can support them at each stage.

To address this gap, we designed a two-week workshop that taught children to create educational videos about science concepts

they were interested in. The workshop combined the general process of video creation with project-based learning principles, and we used digital tools to guide children through topic selection, information gathering, scriptwriting, storyboarding, and final video production. The study aimed to investigate two questions:

- RQ1: How do children engage in the educational video creation process?
- RQ2: What challenges do children face at each stage of the creation process (i.e., information gathering, scriptwriting and storyboarding, video shooting, and editing), particularly those related to the tools provided?

Our contributions are two-fold. First, our findings suggest that children showed different patterns of engagement throughout the process: some were self-motivated and explored both science topics and the video creation tools; some were motivated by their interest in the topic and saw the activity as an opportunity to explore their curiosity; some viewed it primarily as a learning assignment; and some were disengaged. Based on these findings, we suggest that appropriately designing and adopting educational video creation activities in classrooms or informal settings may serve as a dual learning process for children, fostering both science literacy and media literacy. Second, based on the challenges we observed across different phases of the workshop, we aim to extend Shneiderman's CRCD (Collect-Relate-Create-Donate) creativity support tools framework [74, 75] to focus on and better support children's creativity and age-appropriate needs.

2 Related Work

2.1 Children and Media Literacy: Consuming and Prosuming Media

Media literacy is regarded as a critical skill for children to develop in the 21st century [15, 51, 78]. With children gaining increasing access to the Internet and social media, it is important to teach children to engage critically with the media [54]. Although media literacy often overlaps with digital and information literacy, it is broadly defined as the ability to "*access, understand, evaluate, and create communications across various forms*" [6]. In the context of new media, media literacy can be understood as a continuum, ranging from "*consuming*" to "*prosuming*" media [14, 50]. "*Consuming*" refers to the ability to access and engage with media content, as well as use media with varying levels of proficiency. "*Prosuming*" involves the ability to both consume and produce media content using appropriate tools and skills.

Existing research on media literacy has focused on teaching children how to consume media critically. Studies have explored interventions aimed at helping young people recognize misinformation and understand ethical considerations about digital content. For example, a study by Hartwig et al. [35] suggests gaps in adolescents' ability to discern misinformation on social media, highlighting the need for educational efforts to improve media consumption skills. Similarly, Freed et al. [26] emphasize the digital threats that young users face, such as misinformation, underscoring the importance of safe and informed media consumption practices.

While these studies provide valuable insights, they largely address the consumption aspect of media literacy, leaving the production dimension less explored. Recent research has begun to investigate how children engage in prosuming behaviors, using online media to tell stories, express opinions, and create instructional or performative content [13, 84]. For instance, children employ strategies to engage imagined audiences [55] and can create meaningful media, such as instructional videos, to share knowledge and experiences [13]. However, studies also identify challenges in supporting young content creators, including navigating privacy concerns, deciding what content is appropriate to share, and understanding how to engage audiences effectively [29, 58, 84]. Peer and family support, as well as educational and digital interventions, are crucial for enhancing children's ability to create high-quality, meaningful content [13, 55, 56]. These findings suggest that while children have the potential to create and share meaningful media content, structured support or guidance is essential to help them navigate the complexities of media production. Video creation, as a specific form of prosuming, offers a particularly rich context for exploring these challenges and opportunities.

2.2 Children's Video Creation Practices

Children today are creating videos more than ever, both in and out of classrooms, with these videos taking on diverse meanings [40, 42]. Some serve as a form of "*curatorship*" of their lives, preserving meaningful memories to share with others [55, 64]. Others showcase artistic expression, creative storytelling, and play. Many videos, such as instructional content teaching about science, experiments, or small how-to tasks, hold educational value [13], while some children use videos to voice issues that matter to them, making themselves heard in public spheres, challenging dominant narratives, and defining their identities and perspectives on their terms [20, 25, 27, 71].

Two key issues emerge when examining children's media production. The first is audience—how children view and address the audience for their videos. Children often aim to inform their viewers, and their anticipation of the audience varies. Often, they imagine the audience as familiar individuals, such as classmates, teachers, or family members. In other cases, when posting online, they frequently anticipate a broader or "*vast*" audience, unfamiliar individuals they cannot fully know or define [13, 55, 62]. The second key issue is language. Videos allow creators to utilize multimodal media languages, combining verbal and audio-visual elements to in order to create meanings and to communicate with one another [66].

From an interaction design perspective, we want to understand how tools can support video creation practices. This study incorporated various tools to assist students at different stages of instructional video production, including information searching, scripting, and video editing. This approach aimed to identify students' challenges at each stage and explore ways to design tools that better support their creative and educational processes.

2.3 Learning by Teaching Peers, Agents, and Through Videos

In this study, we focus on instructional videos, not only because they are a common type of video children share online but also because

they hold educational value. They provide educational value for audiences who watch them learn about the content, and for the children creating these videos, the act of explaining and teaching the content facilitates learning, as supported by the learning-by-teaching theory [17, 18].

Learning by teaching has long been recognized as an effective method for improving students' cognitive and motivational outcomes. This approach leverages the dual role of teacher and learner, which fosters a deeper understanding of the material through activities like explaining and questioning [17, 18]. This approach engages cognitive processes such as retrieval practice, generative processing, and social presence, fostering deeper understanding and retention [45]. Research identifies two stages in the learning-by-teaching process: preparing to teach and actual teaching [23, 80]. While preparation helps learners digest the information, teaching is equally important, as it requires learners to retrieve information and interact with an audience. With advancements in artificial intelligence, teachable agents have emerged as a powerful tool to enhance learning by teaching. Teachable agents, such as Betty's Brain, simulate students to teach AI agents by building concept maps and receiving feedback from AI [11, 41, 47, 48].

Teaching an imagined audience through video is another form of learning by teaching that offers unique benefits [39, 69, 70]. Unlike live teaching, video creation reduces social anxieties, enabling learners to focus on organizing and explaining content effectively. This method achieves an optimal level of social presence, balancing self-teaching and teaching to a live audience [80]. Research demonstrates that video teaching fosters generative processing, leading to higher-quality explanations and greater cognitive engagement [39].

3 Workshop Design

In designing our workshop, we aimed to explore the feasibility of instructional video creation as a learning experience for children and to understand the processes they engage in to produce these videos. The workshop was structured around project-based learning (PBL) principles and integrated practices specific to video creation. PBL, an inquiry-based and student-centered approach, immerses students in real-world questions and problems, encouraging them to pose authentic questions, gather information, and work toward solutions. A defining element of PBL is the creation of a "*concrete artifact*" that represents students' understanding, insights, and attitudes about the topic [37]. In this context, video creation is a form of PBL where students choose a topic, gather information, and produce an instructional video as their final product, focusing on media skills such as scripting, storyboarding, video shooting, and video editing [22, 82].

In addition, we sought to examine how various tools could support this process and to identify design implications for future tool development. For each session, we integrated existing tools to support the activities, and Table 1 outlines the focus, content, activity, and technology used for each day.

3.1 Stage 1: Choosing a Topic (Day 1)

In Stage 1 of the workshop, we focused on helping students choose a science topic for their instructional video. We began by presenting

Day	Stage	Lecture Content & Activities	Tools Used
1	Choosing a Topic	Lecture Content: Overview of the workshop and examples of instructional videos on YouTube. Activity: Students brainstorm, discuss, and select a science topic individually or in pairs. We provided example questions based on the National Science Standards Guideline for students as inspirations.	None
2	Gathering Information	Lecture Content: (1) Introduction to what information to collect, search engines, keywords to use to search, and some basics of assessing the credibility of information. (2) Introduction to generative AI tools and the negatives and positives of using generative AI tools to gather information. Activity: Students use Google search, kiddle, and researcher designed ai based on playlab.ai to find information on the chosen topic.	Google search, Kiddle, Playlab.ai
3	Scriptwriting and Storyboarding	Lecture Content: Introduction to video script structure (introduction/hook, main content, conclusion) and analysis of an example video. Activity: Students analyze the structure of an example video and start writing scripts based on the information they found.	Google Doc
4		Lecture Content: Introduction to generative AI tools and the negatives and positives of using generative AI tools for script refinement. Activity: Students choose the AI tool for script writing and refinement optionally.	Playlab.ai
5		Lecture Content: Introduction to storyboarding, using visuals and text to map out video scenes. Activity: Students create storyboards, deciding on the flow of their video and visuals for each scene.	Google Doc, Playlab.ai
6	Video Shooting and Editing	Lecture Content: Introduction to basic recording techniques, including tips for using Chromebooks for video shooting. Activity: Practice delivering scripts and recording videos using Chromebooks.	Web Camera
7		Lecture Content: Introduction to video editing, focusing on cutting and trimming clips. Activity: Use Flixier to cut and organize video footage.	Flixier
8		Lecture Content: Instruction on adding multimedia elements, such as text, music, images, and subtitles to videos. Activity: Use Flixier to add multimedia elements, refine, and finalize the video.	Flixier

Table 1: Video Creation Workshop Schedule

an overview of the workshop structure and objectives, emphasizing that the project would involve creating a science instructional video. To foster engagement, we encouraged students to reflect on anything they had learned in school science classes or encountered elsewhere that they were curious about. We invited them to share these interests with the group to start the session with an open discussion on diverse science topics. Next, we provided examples of inquiry-based science questions based on the Next Generation Science Standards (NGSS) guidelines for grades 4 through 7. These examples helped frame the brainstorming process within a school learning context [19]. For instance, we included questions such as *"Why do seasons change throughout the year?"* and *"What adaptations help animals survive in extreme environments?"* to introduce Earth and life science concepts.

These examples were used to facilitate discussions, encouraging students to consider questions they were particularly curious about and might be interested in exploring further. In groups, students discussed which questions they found most intriguing and which they felt enthusiastic about teaching others. After the discussion, each student received a worksheet to help them articulate their chosen question. On this worksheet, they noted the question they were most excited to explore and provided a brief rationale for their selection.

3.2 Stage 2: Gathering information (Day 2)

Stage 2 of the workshop focuses on gathering the necessary information for the videos, including facts, examples, and multimedia images to help students answer their proposed questions. In this session, children were introduced to two tools for finding information: general search engines and conversational search tools.

Prior studies have examined children's information behaviors when using search engines. One such study focused on Yahoo!igans, a search engine powered by Yahoo that was designed to be a safe, age-appropriate tool for kids [8]. Several studies have shown that children can generally find relevant and appropriate information using these systems. However, they often require guidance on using search terms effectively, navigating websites, assessing sources' credibility, and synthesizing the information they find [7, 9]. Despite its strengths, Yahoo!igans faced challenges in its website design, with unclear hyperlinks and confusing images, which made it difficult for kids to use them efficiently [63]. Unfortunately, Yahoo!igans was shut down in 2013 due to a decline in usage. In place of Yahoo!igans, we have chosen Kiddle, a visual search engine specifically designed for kids [2] (Figure 2a). Google's SafeSearch feature powers Kiddle and provides an intuitive, safe search experience. The tool's visually appealing layout is designed to be user-friendly for younger users, presenting age-appropriate and easy-to-navigate results [38].

In addition to traditional search engines, there is growing interest in conversational-based informational searching through smart speakers like Google Home and Alexa. Studies show that about a quarter of children ask questions related to science and technology using these devices. However, these tools still face challenges in providing age-appropriate responses, understanding the context of questions, adapting answers based on the query, and ensuring the accuracy and presentation of the information [52, 53]. We are also

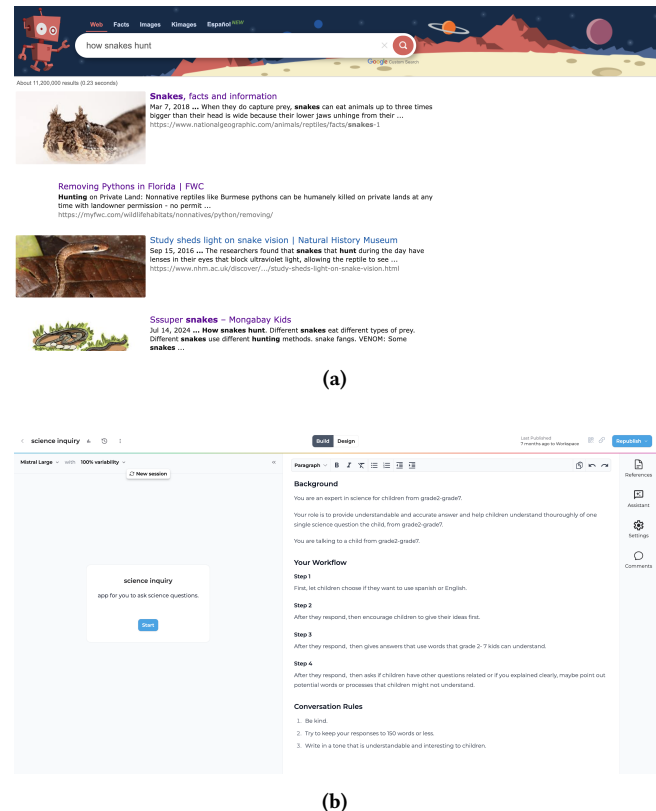


Figure 2: (a) The Kiddle search interface displaying results for the query "How snakes hunt." (b) The Playlab.ai interface allowing researchers to select an LLM model and test prompts.

introducing PlayLab.ai into the workshop to address some of these limitations [3] (Figure 2b). PlayLab.ai is a sandbox platform that allows students to interact with pen-sourced LLM models explicitly designed for educational contexts [57]. Our research team carefully crafted and iterated the prompts used in PlayLab.ai to ensure the outputs are age-appropriate and accurate.

The workshop is divided into two parts. In the first part, we introduced the two search engines (Kiddle and PlayLab.ai) and gave students tips on searching effectively. We covered the basics of using keywords, navigating websites, and assessing the credibility of sources [7, 9]. Additionally, we introduced Generative AI, explaining how it works and discussing the potential for misinformation [34]. In the second part of the workshop, students used the three tools—Kiddle, PlayLab.ai, and their search techniques—to find at least three facts related to their question. They searched for three images that help illustrate these facts and identified at least one real-world example that connects with their research. By the end of the session, students should have gathered various resources and gained a deeper understanding of how to search for, evaluate, and synthesize credible information online.

3.3 Stage 3: Scriptwriting and Storyboarding (Day 3-5)

The scriptwriting and storyboarding activities were conducted over three days. The first two days focused on writing a script with a clear structure for the video using the information gathered in Part 2. The third day was dedicated to organizing the video by deciding on shots and where to use multimedia elements like pictures and sounds. For scriptwriting, we started with a brief introduction to what scriptwriting is and what is typically included in a video script. We explained the structure based on prior works on instructional video frameworks, which include elements like a greeting, main content, engagement strategies (e.g., Q&A, examples), reflection, and conclusion [13, 61, 83]. Then, we presented children with an educational video about photosynthesis on YouTube and asked them to analyze its structure and engagement. For storyboarding, we gave a short lecture covering the basics of visually planning a video. This included explaining how to outline scenes, annotate multimedia elements, and align visuals with the script. We also showed examples of effectively incorporating images, text, and sounds into the storyboard to enhance the narrative.

Similar to stage 2, we used Playlab.ai, an education-focused and customizable LLM tool, to support scriptwriting and storyboarding. Prior research has demonstrated the use of similar AI tools for children's writing tasks [33] and video script and storyboard generation for newsreels [81]. Using insights from these studies, we scaffolded the process to help children write scripts with a clear introduction, main content, and conclusion, incorporating one example and a Q&A segment for audience engagement.

3.4 Stage 4: Video Shooting and Editing (Day 6-8)

During Days 6–8, children focused on video shooting and editing. We began with a lecture introducing basic recording techniques, including tips for using Chromebooks with a video webcam. The lecture covered key aspects such as framing, lighting, and ensuring clear audio quality. Afterward, we introduced children to the editing tool Flixier and demonstrated simple editing techniques [1], such as cutting out mistakes, adding transitions, and refining video sequences. We chose Flixier because it is a user-friendly video editing tool designed for beginners, making it an ideal choice for users with little or no editing experience. Flixier offers an intuitive interface and essential features that allow users to perform basic edits without overwhelming them with complex options. Furthermore, its compatibility with Chromebooks and seamless integration of multimedia elements made it well-suited for the workshop.

4 Study Method

4.1 Study Process

The study was conducted at a local summer camp in a bilingual community (Spanish and English) in Southwestern United States, as part of a non-profit center's educational and social initiatives for the local community. The center offers full-day, out-of-school STEM programs during the summer and other school breaks, allowing children from underserved communities to continue learning in a supportive environment. The camp focuses on exploring Science,

Technology, Engineering, Art, and Mathematics (STEAM) subjects, with classes designed to enhance children's learning experiences outside the traditional classroom. The center relies on grant funding to make the program accessible, ensuring it is free or low-cost for participants.

The camp program spans nine weeks, with one-hour classes every morning. This study was integrated into the summer camp as a two-week workshop on educational video creation. Each class within the workshop followed a consistent structure: a 10-15 minute introduction to the day's content, followed by one or two hands-on activities. These activities build on each other, each session contributing to the final video.

Before the start of the program, students were informed about the data collection process. Consent forms were distributed with the assistance of the community center staff prior to the summer camp and collected on the first day. In addition, oral assent was obtained from the students immediately before the camp began on the first day.

Throughout the workshop, one researcher served as the primary facilitator, delivering lecture content, assisting children, and monitoring their use of AI and other technologies during hands-on activities. The second researcher acted as the primary observer, documenting students' engagement and challenges, while also stepping in to help facilitate the class when needed. Additionally, three teachers from the program supported the workshop by assisting with classroom facilitation and providing help to students as needed.

4.2 Participants

Children in this study were between the ages of 8 and 13. Although more children were enrolled in the class, not all attended the entire two-week program. As a result, only those who attended all sessions were included in the analysis. This led to a final sample of 12 participants from the 4th to 7th grade. Each participant was assigned a unique identifier (P1–P12) to ensure confidentiality. Detailed participant information is provided in the table below (Table 2).

4.3 Data Collection

To create a natural and comfortable learning environment for the children, we avoided using audio and video recording equipment in the workshop. Instead, data collection was integrated seamlessly into the workshop process, with one facilitator leading and moderating the session and an observer taking notes. Following each session, the facilitator and observer held a brief recap discussion to consolidate their observations. This approach ensured that data collection was non-intrusive and aligned with the primary goal of fostering children's learning [65].

We collected two primary types of data during the workshop: observation notes and participant surveys with brief follow-up interviews.

The primary observer took notes with two main goals: (1) documenting students' engagement with the activities and (2) identifying challenges students faced. To structure the observations related to engagement, we used an adapted version of Reeve et al.'s [67] Observer Rating of Student Engagement scale. The scale includes 7-point ratings of attention, cognitive effort, verbal engagement

PID	Grade Level	Gender	Video Topic
P1	6th grade	Male	How energy transfers
P2	7th Grade	Female	Human bones
P3	5th Grade	Male	How snakes hunt
P4	4th Grade	Female	Why cats have furs
P5	5th Grade	Male	Natural selections and adaptations
P6	5th Grade	Female	What would happen if we entered a black hole
P7	5th Grade	Female	What would happen if we entered a black hole
P8	4th Grade	Male	Gravity in games and in real life
P9	4th Grade	Female	How does the human eye work
P10	5th Grade	Female	How does the human eye work
P11	4th Grade	Prefer not to say	Gravity in games and in real life
P12	5th Grade	Female	How do fossils form

Table 2: Information of Participants In the Study

(whether salient or not), and persistence in facing challenges. Given the exploratory nature of the study, the observer did not receive formal training in using the scale or conducting reliability checks within this specific context, so the ratings may be subjective. To address this, the observer and facilitator orally described and discussed reasons to support the ratings and wrote notes describing specific students behaviors explaining the judgments and describing specific student behaviors. Observations were recorded using paper and laptops. Before each session, the observer and facilitator aligned on specific focus areas of the session, and after the session, they discussed and reconciled their notes to ensure consistency.

The other primary data we collected were open-ended surveys from students at the end of each session, where they wrote or drew what they liked or disliked about the activities and technologies. After collecting the surveys from each student, we conducted short, informal interviews with them to clarify their responses or explore their experiences in more depth.

In addition to the primary data, we collected two types of supplementary data to enrich the analysis. This included log data from participant interactions with the technologies (i.e. AI tool logs in Parts 1 and 2). We also collected artifacts produced by participants, including scripts, unedited videos, and edited videos from Parts 2 and 3 of the workshop.

4.4 Data Analysis

To analyze RQ1 (children’s engagement), we synthesized each student’s behavior into a 3-point ordinal engagement score (low, moderate, high) for each of the four activity stages: information gathering, scriptwriting, video recording, and video editing. These scores were derived through qualitative coding based on the observer notes and the engagement rating scale. For example, a student who actively asked questions, revised content independently, and explored video editing features was coded as highly engaged. In contrast, students who gave up quickly when facing challenges, became distracted or off-task, or demonstrated minimal effort in completing tasks were coded as having low engagement. We then visualized these patterns using a heatmap to highlight variation across students and stages. In addition to the visual summary, we

provide a qualitative narrative that contextualizes each engagement profile using quotes and behavioral examples drawn from the observation notes.

To analyze RQ2 (children’s challenges), we adopted an inductive thematic analysis approach [77], reviewing observation notes, survey responses, and informal interview logs to code themes and patterns to emerge organically without imposing preconceived categories. Supplementary materials, including AI tool logs and student-generated artifacts (e.g., scripts and videos), were used to triangulate findings and deepen interpretation of how children engaged with the tools and processes over time.

5 Results

5.1 RQ1: Children’s Engagement Patterns in Video Creation

Students showed different patterns of engagement throughout the four phases of the video creation process: information gathering, scriptwriting, video shooting, and editing (See Figure 3). We identified four types of engagement: self-directed participation, topic-motivated learning, task-oriented learning, and disengaged participation.

5.1.1 Type 1: Self-Directed Participation. Students in this group (P6, P7, P9, P10) showed high engagement throughout all stages of the video creation process. As a note, these four students worked in pairs, and collaboration seemed to support greater persistence and creativity. They stayed focused throughout the class sessions, often going beyond what the facilitator asked. For example, they explored additional features of the video editor or continued to revise their scripts even after class ended. When encountering challenges, they tended to discuss with their partner and troubleshoot independently.

During information gathering, they frequently switched between tabs to search for relevant information and looked up unfamiliar terms to deepen their understanding. For example, P6 and P7 explored vocabulary like “spaghettification” and asked imaginative questions to the AI, such as “What would happen if I entered a black hole?” They actively discussed with each other, requiring little or no facilitator assistance.

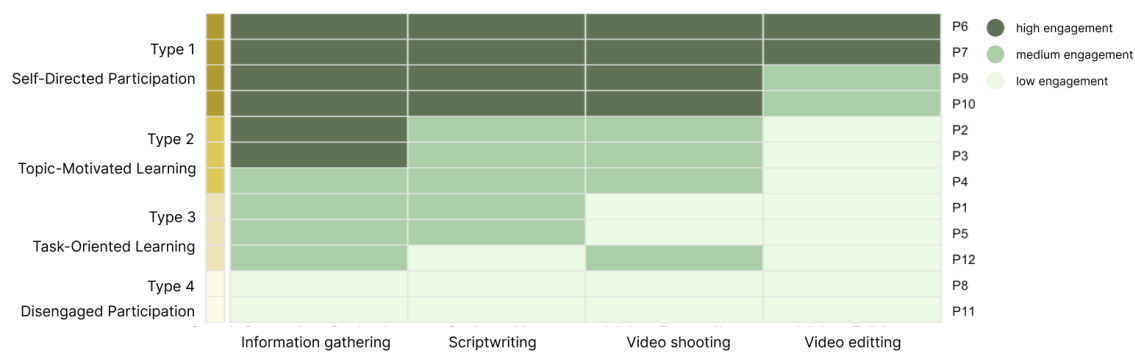


Figure 3: Children's Engagement Levels Across Video Creation Stages.

In scriptwriting, they used the AI tool as a source of inspiration rather than a shortcut. P6, for example, rewrote the content in her own words and added humor to explain difficult concepts. She described spaghettification as *"It's a process that you go through when in a black hole. I know, the name is pretty funny for something that's painful but it's literally in the name. So spaghettification is when the gravitational force of the black hole compresses you from head to toe while stretching you at the same time, so it's basically turning you into a noodle or a ribbon."* P9 and P10 similarly selected useful parts of the AI's output and rephrased them in their own words.

During recording and editing, they divided the script and each person took responsibility for specific parts, and they explored different features in video editing tool. They started with basic features like cutting and rearranging clips, and then experimented with features like auto-transcription, adding images, and subtitles.

5.1.2 Type 2: Topic-Motivated Learning with Decreasing Video Creation Engagement. Students in this group (P2, P3, P4) showed initial engagement during the early stages of the project but gradually disengaged for different reasons. They were initially interested in the topics they chose, and during the information gathering stage, they were curious and motivated to learn more. For example, P2 wanted to know about human bones and searched *"biggest human bones"* in Google, browsing through several pages. P3, though he found search engines difficult to use, engaged with the AI by asking questions and follow-up questions to understand different strategies of how snakes hunt, such as *"venom"* and *"constriction."* Similarly, P4 asked the AI, *"Why do cats have a lot of fur?"* and followed the bot's suggestions closely.

However, their engagement declined during scriptwriting stage, as they found it challenging to organize information into coherent scripts and started to rely on AI for content generation. During the video editing phase, they showed varying levels of interest, but all began to not as engaged. For P2 and P3, this was because they were more interested in learning about the topic than in video editing. P3 said he was *"not really interested in making videos"* and found it *"embarrassing to show myself to others."* Still, he found value in the entire video creation task, saying *"It's not something I really enjoy, but I did learn some things about snakes, how they hunt, I didn't know about that."* P4, on the other hand, was interested in

the activity but found the interface too hard to use. She expressed frustration, calling the system *"wrong,"* and said that it was *"too difficult."* However, when asked if she would like to make videos again, she responded positively, *"Yes, it's interesting, just too much work."*

5.1.3 Type 3: Task-Oriented with Limited Self-Direction. Students in this group (P1, P5, P12) often treated the activity as a structured classroom task, rather than a space for creative or expressive work. During the information gathering stage, they used the search engine and AI tool to find the necessary facts, but typically moved on once they had collected what they felt was sufficient. For example, P1 asked the facilitator if his answer was okay and after receiving confirmation, handed it in without further exploration. P12 showed initial interest but became less engaged when encountering challenges during searching with google search engine.

In scriptwriting, they relied on AI tools to help complete the task, often using the generated text with little revision. For example, P12 used the AI to finish writing after encountering difficulty, and P5 completed the task with minimal revisions. During video recording and editing, they showed a willingness to complete the required tasks, though their performance tended to remain minimal. For example, during editing, most students focused on basic trimming or followed facilitator suggestions. Some found the interface challenging to navigate. P1, for instance, tried importing video clips but described the tool as *"too complicated."*

5.1.4 Type 4: Reluctant or Disengaged Participation. Students in this profile (P8, P11) were neither interested in creating videos nor motivated to learn about the science topics involved. Instead, they preferred activities like gaming, and the provided tools and technologies were not engaging enough to sustain their attention. During information gathering and scriptwriting, they relied heavily on the teacher for help. For example, P11 repeatedly asked for the facilitator's assistance without attempting to search or write independently and shifted to playing games once he believed the task was complete. During video editing, they engaged minimally with the tool unless prompted and expressed that the experience was *"not interesting."* Their final videos were short, consisting primarily of a plain reading of the AI-generated script, with little editing.

5.2 RQ2-1: Challenges in Gathering Information

For gathering information, we identified three major challenges children faced: text-heavy search results that were difficult to digest, difficulties in forming effective search queries, and overtrust in conversational AI search tools.

5.2.1 Challenge 1: Unsuitability of Standard Search Engines. Standard search engines (i.e., Google Search) presented barriers for children due to their text-heavy results and lack of visual aids. Many children found the abundance of text overwhelming, making it challenging to stay focused and engaged. For example, P3 shared, *“There’s too much reading, and I don’t understand everything.”* This sentiment was echoed by others who expressed boredom and frustration when faced with sifting through large amounts of text to find relevant information. Additionally, children strongly preferred visual aids, such as pictures, to help them understand concepts more easily before delving into detailed text. P9 and P10 stated, *“I think it’s better to have pictures. Pictures are easy to see.”* The lack of visual elements in search results made it harder for children to grasp a topic’s overall context, mainly when unfamiliar with specific terms or concepts. Another challenge was the difficulty of choosing, locating, and identifying important information from lengthy text and the vast array of results provided. For instance, when P1 searched for information about how energy transfers, the first website result contained a long explanation about various types of energy. P1 struggled to understand the big picture before diving into the details. Upon encountering unfamiliar terms within the article, P1 felt compelled to open new pages to search for definitions, further complicating and lengthening the process. He described the experience as frustrating and, like P3, remarked that *“it’s too much reading, like schoolwork.”*

5.2.2 Challenge 2: Limitations of Child-Friendly Search Engines. Child-friendly search engines were introduced to provide a safer and more accessible experience for children. These tools often filter results to include more trusted sources than traditional search engines, such as sources National Geographic [32] or sources created explicitly for kids like Mongabay Kids, which are specifically designed for children. They use more straightforward language and include pictures to help illustrate concepts, making information easier to understand. However, despite these advantages, child-friendly search engines also pose challenges. Due to strict filtering or a lack of sophisticated algorithms, these tools sometimes fail to deliver relevant results. For instance, P3 shared, *“I don’t like this one. I searched ‘how snakes hunt,’ and it gave me things about snakes but not how they hunt.”* This disconnect between search intent and the results provided led to confusion.

5.2.3 Challenge 3: Reliance on and Distraction by the Conversational Search Tool. Children generally appreciated AI-powered tools as search engines over traditional search engines, as these tools provided relevant answers to specific questions and allowed for interactivity, such as asking follow-up questions. For example, when using the AI tool, P5 initially asked, *“What is natural selection and adaptation?”* After the bot explained these topics, he asked for examples, which the AI delivered clearly and concisely. Children also valued the simplified and direct explanations provided by the AI. P9 remarked, *“I like the AI tool the most because it talks to you like a*

teacher and gives answers to your questions.” However, children did not fully recognize the potential for AI to produce inaccurate or misleading results. When asked if they thought the AI-generated content was accurate, P9 responded, *“I think so. I don’t know if AI can give wrong answers.”*

Additionally, some children became distracted by the AI, diverting their focus from the task. For example, P8 and P11, tasked with understanding gravity in games and real life, became sidetracked and began asking the AI unrelated questions about the game and popular influencers who play it. This tendency to explore unrelated topics highlighted the challenge of using an open-ended tool, as the AI’s broad capabilities sometimes reduced children’s focus on the actual task. Furthermore, many children relied heavily on AI-generated content, often copying and pasting answers directly without additional effort or synthesis. P12 expressed this sentiment, saying, *“The AI is good enough that I don’t need to look for anything else.”*

5.3 RQ2-2: Challenges in Scriptwriting and Storyboarding

In the scriptwriting and storyboarding phase, we identified three major challenges children faced: difficulties brainstorming and structuring content, declining engagement due to perceived effort and lack of personal interest, and reliance on tools during the creation process.

5.3.1 Challenge 1: Brainstorming and Structuring Content. Many children struggled to structure their scripts effectively, including creating a proper outline with a hook, main content, and conclusion to present their ideas. They also faced challenges in organizing information into a coherent sequence of frames. For example, P3 shared, *“I know about the three ways snakes kill prey, but I don’t know how to write a script.”* On the first day of scripting and storyboarding without AI support, most children (11 out of 13) got stuck on the main content. They found it difficult to independently translate the information they had gathered in the previous step into something clear and understandable for their audience. Additionally, children faced challenges brainstorming creative ways to convey the content, such as role-playing, acting, or incorporating different materials like drawings or physical objects. Instead, all of them chose to narrate the concepts.

5.3.2 Challenge 2: Loss of Interest in Scriptwriting. The scripting and storyboarding phase marked a turning point where many children, particularly those in Group 4, began to lose interest. During the scriptwriting stage, several students struggled to engage with the task. Many stared at blank pages for extended periods, unsure of how to begin, while others frequently sought help from the teacher, asking questions such as, *“How do I write a script?”* A few students shifted their focus entirely, using the Chromebook to play games instead of focusing on the activity. Two reasons for their disengagement emerged from observations. First, while some students appreciated the structure that scripting provided, others found the process tedious and unnecessary. For example, after a short lecture on analyzing an instructional video on YouTube and covering some basics about scripts, P5 remarked, *“Why write a script? I already*

know about it. I just want to talk and record. It's faster." Additionally, several students (P3, P5, P12) expressed frustration with the effort required to create a script. One child candidly stated, *"It's too difficult. You have to think so much, and I don't know where to start."*

5.3.3 Challenge 3: Reliance on AI Tools. While most students, including those who initially struggled to translate their ideas into scripts and storyboards, successfully used the AI tool to produce scripts with clear introductions, main points, and conclusions, the tool's design played a crucial role. It guided students step-by-step in writing their scripts; however, when students faced challenges, the tool often provided completed outputs. While this approach ensured task completion, it ultimately made students overly dependent on the tool's output. Most children (9 out of 13) opted to copy the AI-generated content with minimal adaptation directly. For example, P3 shared, *"I liked the AI writing because it helped me,"* and P2 stated, *"It helped me on what to do with the script."* While students appreciated the assistance, the resulting outputs often lacked personalization, self-expression, and creativity. Instead, the scripts followed generic patterns, heavily influenced by the AI's tone and style. This reliance on AI-generated content undermined the core goal of empowering youth to create media that amplifies their unique voices in the public sphere. Rather than showcasing their individuality, the outputs reflected the AI's writing style, limiting opportunities for authentic self-expression and creativity.

5.4 RQ2-3 Challenges in Video Shooting and Editing

In the video recording and editing phases, children faced two major challenges: discomfort with recording and sharing videos, and difficulties navigating the complexity of the video editing system.

5.4.1 Challenge 1: Discomfort with recording and sharing videos. Some children were willing to use the webcam to record videos of themselves reading directly from their script, engaging with the process without hesitation. Others, however, showed reluctance and discomfort when it came to recording. For example, P1 explained, *"It's embarrassing to talk to the camera,"* describing the uneasiness that arose from speaking directly into the lens. In addition to the challenges of recording, some children expressed concerns about the thought of others watching their videos. P8 said, *"I don't want to record. I don't want other people to see me,"* suggesting the apprehension associated with the possibility of their work being viewed by others.

5.4.2 Challenge 2: Complex video editing system navigation. Only five students actively engaged in the video editing process, while the majority performed only basic tasks, such as dragging their recorded videos into the editor, resizing them, or merging two clips. Many children did not explore the full range of editing tools due to a lack of interest in learning about video editing and feeling overwhelmed by the system's complexity. P3 stated *"too many buttons, too complicated"* and similarly P12 stated *"I don't know what I'm doing with it"*. For these students, the abundance of options and the perceived difficulty of understanding the tools made the editing process unappealing and challenging.

The children who engaged more deeply with the editing process, following the guidance of the researcher facilitator, were able to

explore and apply a variety of features. They successfully trimmed unwanted parts from their videos, merged clips, added transitions, inserted pictures, and applied auto-transcription. Some students, such as P6 and P7, even experimented with additional features like subtitle effects, finding these creative elements enjoyable.

However, even among the engaged students, several challenges were observed during the video editing process (Figure 4). For instance, resorted to trial and error, clicking on random buttons and undoing actions when the results were not as expected. When it came to trimming and merging clips, students repeatedly played back segments of their videos, pausing and rewinding multiple times to identify precise points for cutting. Dragging and rearranging clips on the timeline presented another challenge, as some struggled to align the clips properly or accidentally overlapped them, requiring further adjustments. Additionally, P6 and P7 showed curiosity about advanced features, such as adding special effects or experimenting with overlays. They asked questions to the facilitator like, *"What else can I do with this?"* or *"How do I make it look cooler?"* However, when they explored beyond the facilitator's guidance, they often became stuck and asked for further guidance.

6 Discussion

6.1 Children's Video Creation as a Learning Activity

6.1.1 Educational Video Creation as a Dual Learning Opportunity for Science Literacy and Media Literacy. We observed that some children were more engaged in understanding scientific content, while others were more interested in the video production process. This variation implies that the creation of instructional videos can serve as a way to integrate science and media literacy, allowing children to develop skills in both areas.

For children interested in learning about science content, the process of searching, organizing, and synthesizing information became a way to prepare for teaching. This aligns with the learning-by-teaching framework, which suggests that preparing and explaining concepts can deepen understanding [23, 80]. Although not all children participated in this process, some attempted to explain complex scientific concepts in their own words and used multimedia representations to convey the content effectively.

From a media literacy perspective, past research suggests that compared to adult-created videos, children's videos tend to have less structure and minimal editing [55]. This workshop introduced a systematic media production process and provided tools for children to create more structured videos. Future studies could further explore how this dual learning process influences children's long-term engagement with science communication and media production, investigating whether children continue creating educational videos beyond the workshop or how they apply media literacy skills in other contexts.

6.1.2 How to Support Children Who Struggle during Video Creation? One issue we observed is that children showed different levels of engagement during the video creation process. Although we did not conduct an experimental test, our observations suggest that only students who were driven both by curiosity in the science topic and persistence in overcoming challenges (Type 1) were able to push

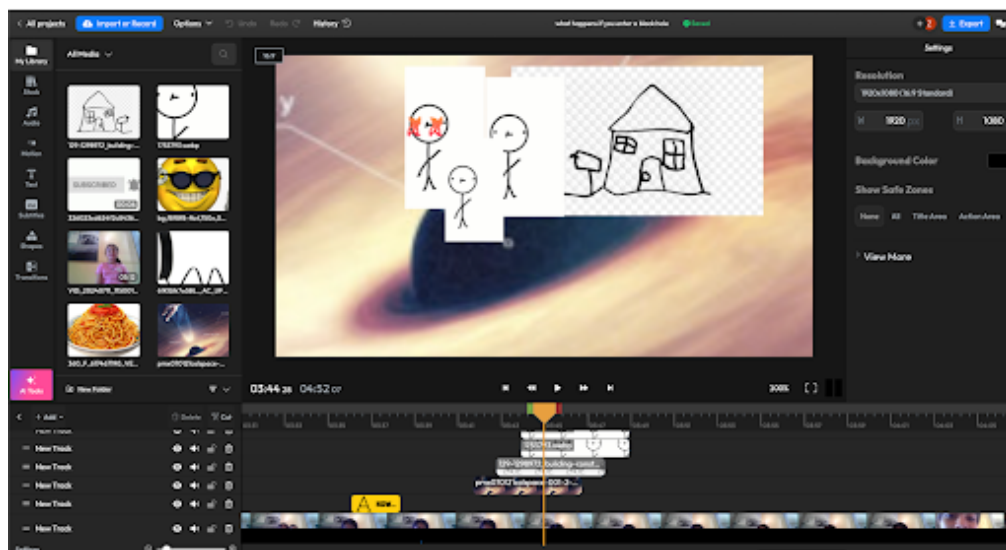


Figure 4: Video editor interface where children worked to create their videos.

themselves and ultimately create original videos with rich content and strong visualizations. In contrast, Type 2 students, who had initial curiosity, often became stuck when facing difficulties; Type 3 students took the task more like a regular assignment and invested less creative effort; and Type 4 students were largely uninterested from the beginning.

This raises an important question: how can we motivate students of different types to engage more deeply, like Type 1 students? While we do not have a clear answer yet, there are three possible directions. First, grouping students may help, as both suggested in prior research and what we observed in our study [44]. When working in pairs or groups, children encourage, question, answer, explain, build on each other's work, learn from one another, and contribute more actively [31]. Second, providing scaffolding or modeling to help students overcome difficulties during the task without promoting overreliance on the tools. For example, instead of offering direct outputs, AI systems could model processes or provide step-by-step guidance [33, 34]. Finally, we need to consider how we might foster intrinsic motivation for video creation itself. Although we initially anticipated that most of the children would be motivated by making videos (as suggested by surveys showing a high percentage of children enjoy watching and sharing videos and aspire to be influencers [60]), this was only true for some participants (e.g., P6, P7, P4). Others were hesitant due to discomfort with sharing or a perceived lack of meaning in the activity. Future work could explore ways to personalize the experience. One thing we observed is that children enjoy playing games and engaging in pretend play. Thus, one possible way might be incorporating these storytelling elements, such as game-based presentations or pretend-play narratives, may better support students who do not initially see value in the task.

6.2 Design Implications for Creativity Support Tools for Children

Our study explored the process of child-led educational video creation — from information searching to scripting and production. Drawing on Shneiderman's Collect-Relate-Create-Donate framework for creativity support tools [74, 75], our findings highlight how this model can be adapted to inform the design of tools that support children's creation (See Table 3).

6.2.1 Collect: Combine Filtered Internet Retrieval, Conversational Search, and Multimodal Results. In Shneiderman's framework [74], Collect refers to "the process of learning from existing works, such as those stored in libraries or on the Web". In our study, this stage corresponds to the information searching phase, where children gathered relevant content for their video topics.

Prior research suggests that children often have difficulties in information searching due to several factors. For example, they may not know what keywords to use or how to adjust their searching strategies after failure, something adults tend to do more effectively [10, 46]. This was also evident in our study, where children expressed a preference for natural language interactions with AI, such as "giving an answer to a question" (P9), without needing to worry about crafting precise keywords.

However, conversational search is not without its challenges. One major concern is that it can generate biased or hallucinated information. While it is true that misinformation can also come from humans or general internet searches, both our findings and prior research show that children are less likely to question or think critically when interacting with AI [34, 43]. This risk becomes more severe when the AI appears all-knowing or when children develop a strong sense of trust in generative AI. Content moderation strategies, such as parental oversight or automated mechanisms like Retrieval-Augmented Generation (RAG) [28], may help address

Stage	Definition	Observed Children's Challenges	Design Directions
Collect	Learn from existing works, such as libraries or the Web.	(1) Overwhelming text-heavy search results. (2) Difficulty forming effective search queries. (3) Overtrust in conversational AI responses.	Support conversational, filtered, and multimodal information gathering.
Relate	Consult with peers and mentors at early, middle, and late stages.	(1) Different engagement interests among students.	Use role-based peer and family collaboration to align interests.
Create	Explore, compose, and evaluate possible solutions.	(1) Difficulty transforming information and ideas into a concrete output. (2) Loss of motivation due to perceived effort and reward. (3) Reliance on tools. (4) Adult-oriented interfaces too complex for children.	Scaffold and model the process of structuring and transforming ideas into outputs; design low-floor, high-ceiling tools balancing support and autonomy.
Donate	Disseminate results to gain recognition and contribute to the community.	(1) Embarrassment showing oneself to others. (2) Privacy concerns.	Encourage meaningful and safe sharing of content; teach children about appropriate content to share and anticipating potential audience feedback.

Table 3: Extension of Shneiderman's Collect-Relate-Create-Donate (CRCD) framework based on observed challenges and design directions in children's educational video creation.

these concerns by combining a retrieval mechanism that pulls information from child-friendly databases to ensure accuracy and appropriateness and a generation component that synthesizes this information into clear, structured responses.

Additionally, children's preferences for search results often differ from those of adults. In our study, many children suggested that the text-heavy responses were overwhelming and not intuitive. They expressed a desire for larger pictures and more visual elements. This aligns with prior research on multimodal reading [59, 72, 73], which suggests that combining written language with images and other modes enhances children's engagement and comprehension [76].

6.2.2 Relate: Support Peer and Family Role-Based Collaborative Creation. Relate refers to "consult with peers and mentors at early, middle, and late stages." [74] Both peers and parents play an important role in supporting children's creativity. Research on children's group creativity highlights how certain patterns of talk—like emotionally engaged conversations and shared dialogue spaces known as the "collaborative floor"—can spark joint idea generation and deeper creative thinking [79]. Parents and peers don't just help children complete creative tasks—they also nurture creative development more broadly by encouraging autonomy [90], offering metacognitive prompts [85], modeling creative behavior ([5] as cited in [4]), and providing supportive scaffolding [21]. In the context of video creation, Cai and Wei's analysis [13] of child-featured YouTube videos shows that parents often play an active role by demonstrating steps, offering help, and asking thoughtful questions to guide children's thinking.

In our observations, we noticed that children showed different interests during the video creation process. Some were more interested in researching and learning about the science topic, while others were excited by video creation and storytelling. To support these differences, creativity support tools could allow for role-based

collaboration, letting children take on roles that match their interests and learn from each other. For example, a child who enjoys gathering information could be the Researcher, while another who loves coming up with stories might be the Scriptwriter. Creating space for flexible, interest-driven roles helps children work together more effectively—and makes the creative process more engaging and meaningful for everyone involved.

6.2.3 Create: Fostering Appropriate Reliance on Tools and Supporting Flexible Composition. In our study, two main types of tools were used during the creation phase, where children "explore, compose, and evaluate possible solutions" [74].

The first type included generative AI tools for scriptwriting. Similar to findings in prior studies on children's use of generative AI, we observed that when AI systems were allowed to generate content based solely on prompt design, some students relied heavily on the AI. Even when prompts attempted to set limitations, students bypassed them (if the system was not designed to make these constraints difficult to override), completing their work with minimal creative engagement [34, 87].

To address the challenge of children's reliance on AI, existing tools have been designed to constrain full AI generation [12]. For example, instead of generating complete scripts, tools are designed to provide guiding questions, hints, or partial prompts to encourage children to generate ideas themselves [16, 43, 89]. In some tools, AI responses are delayed until children first contribute their own input, which the system can then help refine or build upon [16, 88, 89]. Similarly, in video creation context, during the scriptwriting and storyboarding stages of video creation, generative AI can shift from acting as a content generator to serving as a creative facilitator, guiding children through diverse ways to deliver the content, such as small experiments, role-playing, or using visual props like crafts and models [24, 49]. For instance, when a child plans to teach "how snakes hunt," the AI might suggest: "why not act out a snake to

show how it hunts?” or “could you use clay or rope to model a snake’s movement?”

The second category of tools involves video creation technologies. During the recording phase, some children expressed discomfort with being on camera and feeling socially exposed [36]. To reduce this discomfort and protect privacy, video tools could integrate features such as face filters, virtual backgrounds, or avatar overlays [86]. Additionally, children demonstrated varied levels of interest and ability in video editing. Some found current interfaces too complex and struggled to stay engaged, while others were eager to explore and push their creative boundaries. To support this diversity, video editing tools should adopt a low-floor, high-ceiling design approach [68], making it easy for beginners to get started while providing more advanced features for those ready to dive deeper.

6.2.4 Donate: Sharing Meaningfully While Ensuring Safety and Well-being. In Shneiderman’s framework, Donate refers to disseminating the results and contributing to public knowledge, such as sharing with a wider audience or adding to content libraries. In our study, the final step of the video creation process would involve the potential for children to share their educational videos so others can learn from the videos they create. However, we did not explicitly study the sharing or reception of these videos by external audiences, as doing so would have required additional ethical approvals and raised data privacy concerns¹, and could have posed risks to participants. In addition, several children expressed discomfort with publicly posting their videos.

That said, the workshop encouraged children to see themselves not only as consumers of online educational content but also as creators capable of structuring and producing their own videos in a meaningful and information-rich way — potentially to be shared with friends, family, or even broader audiences [13]. To build on this foundation, future workshops could include additional discussions on content sharing, audience awareness, and privacy considerations, which would help children reflect on the ethical and safety aspects of media creation and better prepare them for responsible participation in online spaces [15].

7 Limitations and Future Work

This research has several limitations. First, the sample size was limited, with only 12 participants completing the entire workshop, all from a specific population and academic background. This small sample may not fully capture the diverse range of children’s experiences, motivations, and skill levels in instructional video creation. However, this sample size does not limit our contribution to exploring the usage of AI in supporting children-created videos. Regardless of different engagement patterns among children, we found that AI has supported children in creating instructional videos and learning by teaching. As we have identified several engagement patterns in a small group of children, future studies should involve larger and more diverse participant groups, including children from different educational backgrounds and varying levels of prior experience with video production.

¹For example, YouTube regulations prohibit children under 13 from creating accounts or posting videos without parental consent as per COPPA

Second, while the main goal was to provide a safe and natural environment for learning, we intentionally avoided collecting intrusive data such as detailed demographic surveys (e.g., personalities, prior experience with AI or video editing tools) or audio/video recordings during the workshop sessions. As a result, we could not fully examine how individual differences, such as prior experience or personality traits, may have influenced students’ engagement in the video creation tasks, nor could we capture nuanced behaviors and interactions throughout the process. Instead, we collected observation notes and end-of-day session evaluation surveys, as noted in Section 4.3. The benefit of this approach is that it allowed us to gather data in a more naturalistic and non-intrusive manner, creating a safer and more comfortable space for children to engage in the learning activities. However, this also introduced data limitations. Without rich multimodal data, we could not analyze detailed interaction patterns with the tools, and the absence of comprehensive demographic information limits the generalizability and reliability of our findings. Process data could add insights into the nuanced analysis of engagement. However, this is not the focus and scope of our study. Future studies can collect related process data during video creation to further explain children’s engagement and learning.

Third, the study was conducted within a structured workshop setting in a summer camp context, where children received guided support at each stage of the video creation process. While this approach provided valuable insights into their engagement and challenges, it may not fully reflect how children independently create videos outside of a structured learning environment. Future research should explore children’s video creation behaviors in more naturalistic settings, such as at home or classroom projects with less direct facilitation.

Finally, the tools selected for this study may not fully represent the range of video production tools available to children. The specific tools used may have influenced children’s experiences, limiting the generalizability of the findings. Future research should examine various AI-assisted scriptwriting, video editing, and interactive storytelling tools to assess their usability and impact on children’s creative processes and learning outcomes.

Our study serves as an exploratory first step to understanding the overall process and challenges students face during instructional video creation. Although the findings may not generalize across different child populations, age groups, or types of creative activities, this research contributes to the understanding of children’s engagement in creating instructional videos, interacting with AI tools, and learning by teaching behaviors. Future research could focus on specific aspects, such as comparing the engagement and challenges of introverted versus extroverted children and exploring tailored supports for different learner needs.

8 Conclusion

This study explored children’s engagement and challenges in creating educational videos, highlighting the dual learning benefits of developing both scientific and media literacy. While some children were motivated by content learning, others were drawn to the creative aspects of video production, demonstrating the potential of

video creation as a multifaceted learning experience. However, challenges such as difficulty in information gathering, over-reliance on AI-generated content, discomfort with recording, and complexity in video editing indicate areas where design interventions can better support young creators. Based on our findings, we propose design implications to facilitate structured media production, encourage creative thinking, and accommodate diverse skill levels. Future research should further investigate how these design interventions influence children's engagement with video creation and its impact on their learning.

9 Selection and Participation of Children

We conducted a workshop with children aged 9 to 13 who were enrolled in a summer camp program. Participants were attendees of the camp and were not purposefully selected for this study. Given the sensitive nature of recruiting children, the parents of all participating children were initially contacted to obtain consent for their children's involvement in the study, and children provided oral assent on the first day of the workshop.

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